



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

CHEMISTRY
JC2 Preliminary Examination
Paper 1 Multiple Choice

9729/01
22 Sep 2017
1 hour

Additional Materials: Data Booklet
 Optical Mark Sheet (OMS)

READ THESE INSTRUCTIONS FIRST

On the separate multiple choice OMS given, write your name, subject title and class in the spaces provided.
Shade correctly your FIN/NRIC number.

There are **30** questions in this paper. Answer **all** questions.

For each question there are four possible answers **A, B, C** and **D**.

Choose the one you consider correct and record your choice using a **soft pencil** on the separate OMS.

Each correct answer will score one mark.

A mark will not be deducted for a wrong answer.

You are advised to fill in the OMS as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Any rough working should be done in this question paper.

Answer all questions

- 1** An acidified solution of the salt KClO_x will oxidise $\text{Fe}^{2+}(\text{aq})$ to $\text{Fe}^{3+}(\text{aq})$ quantitatively, the chlorine being reduced to $\text{Cl}^{-}(\text{aq})$.

When 0.302 g of the salt KClO_x was reacted with $0.540 \text{ mol dm}^{-3} \text{ Fe}^{2+}(\text{aq})$ in the presence of $\text{H}^{+}(\text{aq})$, 21.0 cm^3 of $\text{Fe}^{2+}(\text{aq})$ was needed for complete reaction.

Which of the following is the value of x ?

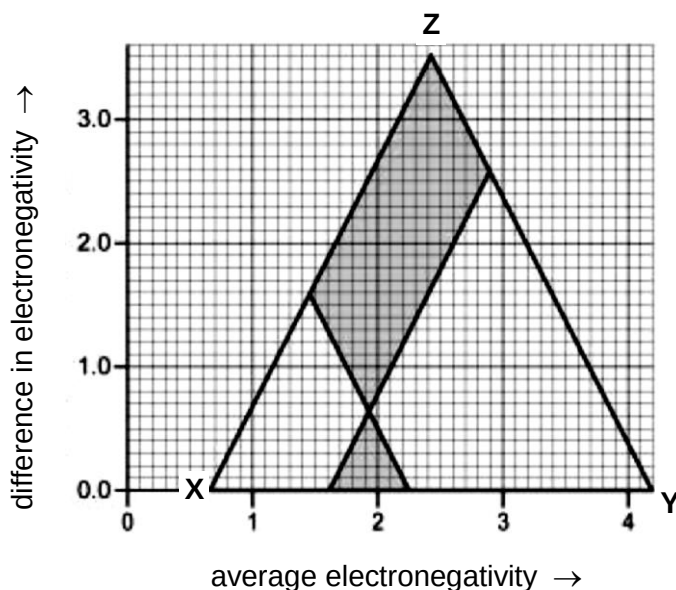
- A** 1
 - B** 2
 - C** 3
 - D** 4
- 2** A student made up a 0.10 mol dm^{-3} solution of $\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$ which she found in the laboratory cupboard and left the solution in an open beaker. A week later, she returned to the laboratory, used the solution for titration with $0.10 \text{ mol dm}^{-3} \text{ HCl}$ and was surprised to discover that her titres were lower than expected.

Which of the following explains why the values were so low?

- A** Some of the barium hydroxide has reacted with carbon dioxide in the air to form solid barium carbonate.
- B** Some water had evaporated from the barium hydroxide solution.
- C** The concentration of HCl was less than the stated 0.10 mol dm^{-3} .
- D** The crystals had less water of crystallisation than stated.

- 3 The type of bonding between two elements can be rationalised and even predicted using a van Arkel triangle. The triangle is based on electronegativity values. Difference in electronegativity is plotted along the y-axis and average electronegativity is plotted along the x-axis.

What is the type of bonding present at each of these bonding extremes, labelled X, Y and Z on the triangle?



- | | X | Y | Z |
|---|----------|----------|----------|
| A | metallic | covalent | ionic |
| B | covalent | metallic | ionic |
| C | covalent | ionic | metallic |
| D | ionic | covalent | metallic |
- 4 Equimolar amounts of aqueous ClO_2 and hydroxide ions react according to the equation below.

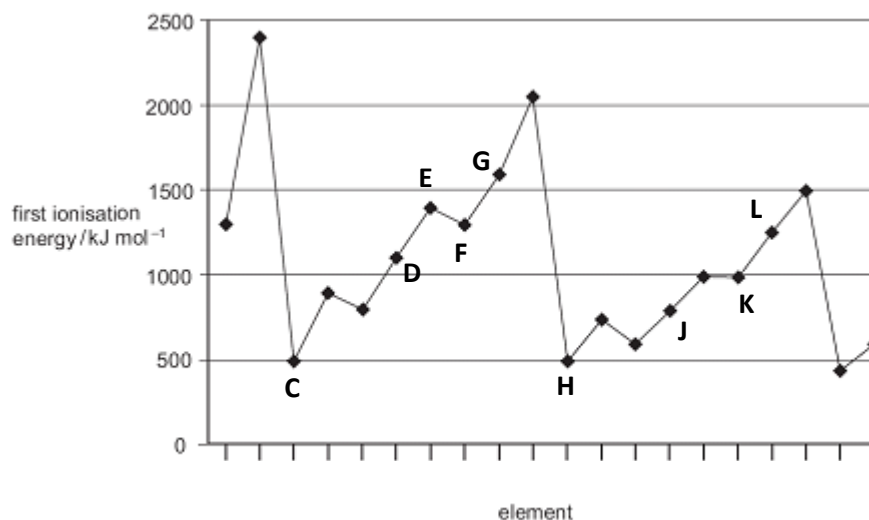


Which statement is correct?

- A The shape changes from linear in ClO_2 to bent in ClO_2^- .
- B The shape changes from bent in ClO_2 to trigonal planar in ClO_3^- .
- C The bond angle changes from about 120° in ClO_2 to 107° in ClO_3^- .
- D The bond angle changes from about 105° in ClO_2 to 107° in ClO_3^- .

5 Use of the Data Booklet is relevant to this question.

The first ionisation energies of successive elements in the Periodic Table are represented in the graph.

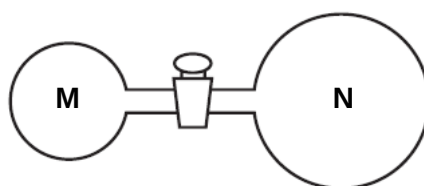


Which of the following statements about elements **C** to **L** are correct?

- 1 **E** and **J** have half-filled p-orbitals.
- 2 Chlorides of **D** and **J** undergo hydrolysis.
- 3 **F** and **K** reacts to form acidic compounds.
- 4 **D** and **F** forms compounds with formula **DF** and **DF₂**.
- 5 **G** and **L** forms compounds with variable oxidation states from -1 to $+7$.

- A 1 and 4 only
 B 2 and 5 only
 C 1, 3 and 4 only
 D 2, 3 and 5 only

6 Two glass vessels **M** and **N** are connected by a closed valve.

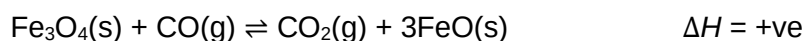


M contains helium at $25\text{ }^{\circ}\text{C}$ at a pressure of $1 \times 10^5\text{ Pa}$. **N** has been evacuated, and has three times the volume of **M**. In an experiment, the valve is opened and the whole set-up placed in boiling water.

What is the final pressure in the system?

- A $3.13 \times 10^4\text{ Pa}$
 B $3.76 \times 10^4\text{ Pa}$
 C $1.00 \times 10^5\text{ Pa}$
 D $1.33 \times 10^5\text{ Pa}$

- 7 Consider the following equilibrium system:



Given that $K_p = K_c (\text{RT})^{\Delta n}$, where Δn is the sum of gaseous products – sum of gaseous reactants.

Which of the following statements are correct?

- 1 $K_c = K_p$ for this system.
- 2 The equilibrium constant, K_c , for this reaction has no units.
- 3 Adding FeO causes the position of equilibrium to shift to the left.
- 4 Increasing the temperature causes the position of equilibrium to shift to the right.

- A 1 and 2 only
 B 1 and 3 only
 C 2, 3 and 4 only
 D 1, 2 and 4 only

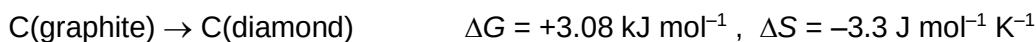
- 8 In an experiment to measure the enthalpy change of neutralisation, 20 cm³ of aqueous sulfuric acid containing 0.02 mol of H₂SO₄ is placed in a plastic cup of negligible heat capacity.

A 20 cm³ sample of aqueous sodium hydroxide containing 0.04 mol of NaOH, at the same initial temperature, is added and the temperature rises by 15 K.

If the heat capacity per unit volume of the final solution is 4.2 J K⁻¹ cm⁻³, what is the enthalpy change of neutralisation?

- A $-\left(\frac{20 \times 4.2 \times 15}{0.04 \times 1000}\right) \text{ kJ mol}^{-1}$
 B $-\left(\frac{40 \times 4.2 \times 15}{0.04 \times 1000}\right) \text{ kJ mol}^{-1}$
 C $-\left(\frac{20 \times 4.2 \times 15}{0.02 \times 1000}\right) \text{ kJ mol}^{-1}$
 D $-\left(\frac{40 \times 4.2 \times 15}{0.02 \times 1000}\right) \text{ kJ mol}^{-1}$

- 9 The conversion of graphite into diamond is a non-spontaneous reaction at 298 K. The entropy change of this reaction is negative at 298 K.

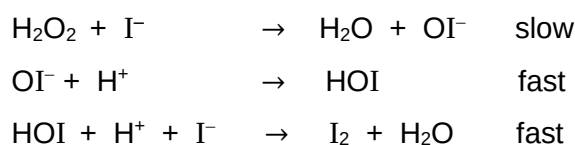


With reference to the enthalpy change, ΔH , of the reaction above, which statements are correct?

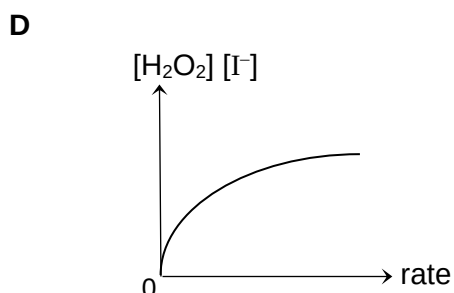
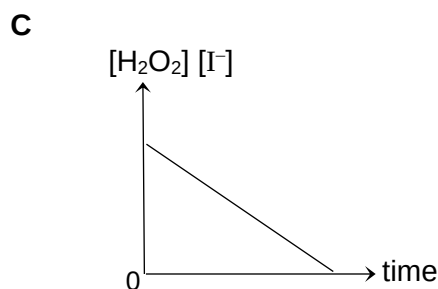
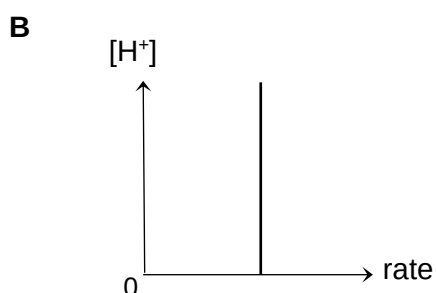
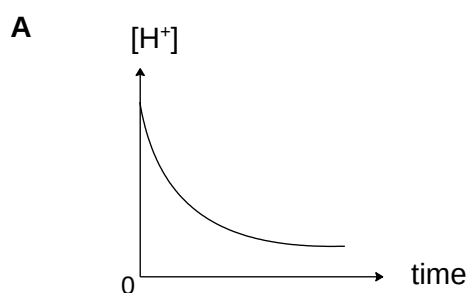
- 1 Enthalpy change of atomisation of diamond is more endothermic than that of graphite.
- 2 Enthalpy change of combustion of diamond is more exothermic than that of graphite.
- 3 Bond energy of carbon-carbon bonds in graphite is greater than that in diamond.
- 4 The activation energy to convert graphite to diamond is lower than the activation energy for the reverse reaction.

- A 1 and 4 only
 B 2 and 3 only
 C 1, 2 and 3 only
 D 2, 3 and 4 only

- 10 The reaction between acidified KI and aqueous H_2O_2 involves the following steps:



Which graph would be obtained?

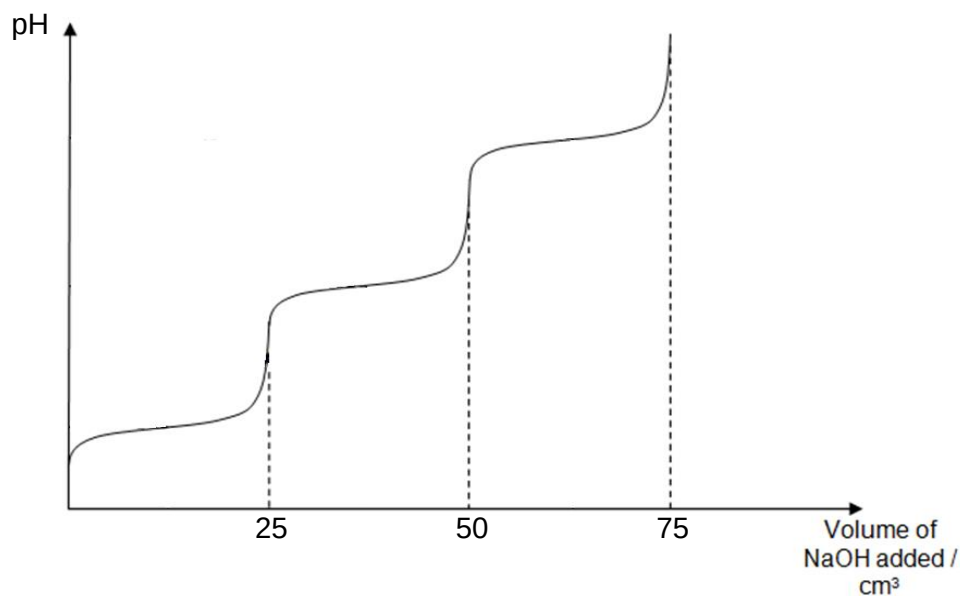


- 11 Compounds of beryllium are amphoteric, like compounds of aluminium.

In which equations is the beryllium-containing reactant acting as a Brønsted-Lowry acid?

- A $\text{BeO(s)} + 2\text{OH}^{\text{-(aq)}} + \text{H}_2\text{O(l)} \rightarrow [\text{Be(OH)}_4]^{2\text{-(aq)}}$
 B $\text{BeCl}_2 + 4\text{H}_2\text{O} \rightarrow \text{Be(H}_2\text{O)}_4^{2+} + 2\text{Cl}^-$
 C $2\text{Be} + \text{O}_2 \rightarrow 2\text{BeO}$
 D $\text{Be(H}_2\text{O)}_2(\text{OH})_2 + 2\text{OH}^- \rightarrow \text{Be(OH)}_4^{2-} + 2\text{H}_2\text{O}$

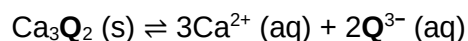
- 12 25 cm³ of 1 mol dm⁻³ of the fully protonated form of compound **L** was titrated against a standard sodium hydroxide solution of a similar concentration and the following titration curve was obtained.



Which of the following compounds is a possible identity of compound **L**?

- A $\text{NH}_2\text{CH}(\text{CH}_2\text{CH}_2\text{CONH}_2)\text{COOH}$
 B $\text{NH}_2\text{CH}(\text{CH}_2\text{CH}_2\text{CH}_2\text{OH})\text{COOH}$
 C $\text{NH}_2\text{CH}(\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2)\text{COOH}$
 D $\text{NH}_2\text{CH}(\text{CH}_2\text{CH}_2\text{CHO})\text{COOH}$

- 13 A sparingly soluble calcium salt dissociates in solution according to the equation:



If the solubility product of Ca_3Q_2 is p , what is the concentration of Q^{3-} at equilibrium in a saturated solution of $\text{Ca}_3\text{Q}_2 (\text{aq})$?

- A $2p^{\frac{1}{5}}$
- B $2\left(\frac{p}{36}\right)^{\frac{1}{5}}$
- C $2\left(\frac{p}{72}\right)^{\frac{1}{5}}$
- D $2\left(\frac{p}{108}\right)^{\frac{1}{5}}$

- 14 A transition metal cation can exist in a 'high spin' or 'low spin' state while exposed to different ligands in an octahedral complex.

A 'high-spin' state has the electrons occupying all the d-orbitals singly first, before pairing in the lower energy d-orbitals.

A 'low-spin' state has the lower energy d-orbitals filled first, pairing up if necessary, before the higher energy d-orbitals are filled.

Which cation would have the largest number of unpaired electrons?

- A 'High-spin' Mn^{3+}
- B 'High-spin' Fe^{3+}
- C 'Low-spin' Co^{2+}
- D 'Low-spin' Ni^{4+}

- 15 Which of the following is **not** a result of the small energy gap between 3d and 4s orbitals?

- A Transition metals have higher electrical conductivity than aluminium.
- B Transition metals can exhibit variable oxidation states.
- C Transition metals have higher melting point than calcium.
- D Transition metals ions are usually coloured.

16 Use of the Data Booklet is relevant to this question.

Two catalysed reactions and their respective catalysts are shown below.

Reaction	Equation	Catalyst
I	$\text{S}_2\text{O}_8^{2-} + 2\text{I}^- \rightarrow 2\text{SO}_4^{2-} + \text{I}_2$	Fe^{3+}
II	$2\text{MnO}_4^- + 5\text{C}_2\text{O}_4^{2-} + 16\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O}$	Mn^{2+}

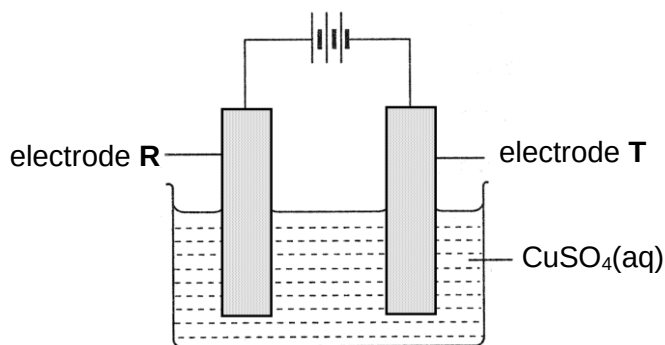
Which statements about the reactions and the catalysts are correct?

- 1 Mn^{2+} is an auto-catalyst in reaction II.
- 2 Fe^{2+} could replace Fe^{3+} as a heterogeneous catalyst in reaction I.
- 3 Mn^{2+} could replace Fe^{3+} as a homogeneous catalyst in reaction I.
- 4 Both catalysts increase the energetic feasibility of their respective reactions.

- A 1 and 3 only
 B 2 and 3 only
 C 1, 2 and 4 only
 D 2, 3 and 4 only

17 Use of the Data Booklet is relevant to this question.

Impure copper obtained from copper ores can be purified by electrolysis as shown below. The cell potential is adjusted such that copper of the impure copper electrode dissolves and impurities such as silver, iron and zinc can be removed through this process.

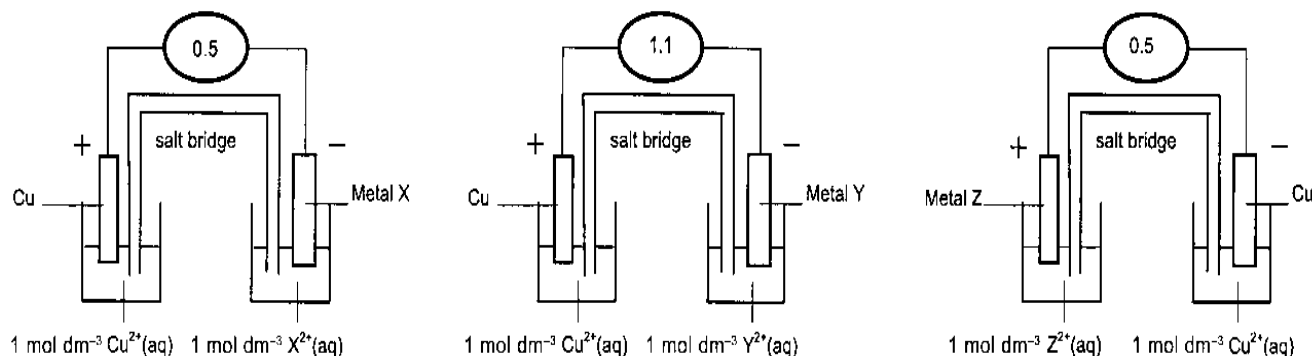


Which statement is **not** correct regarding the above electrolytic process?

- A Reduction occurs at electrode T.
 B Electrode R contains impure copper.
 C Zinc impurity goes into the solution as Zn^{2+} ions.
 D Iron and silver impurities fall to the bottom as sludge.

18 Use of the Data Booklet is relevant to this question.

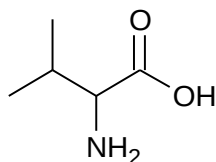
Three electrochemical cells are set up as shown below. The *e.m.f* in volts is shown on each voltmeter.



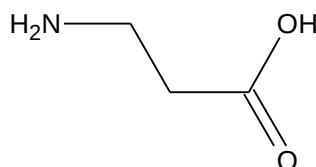
These *e.m.f* indicate the order of reactivity of the metals. The order of the weakest to the strongest reducing agent is

- A X, Y, Cu, Z
- B Cu, Z, X, Y
- C Z, Cu, X, Y
- D Y, X, Z, Cu

19 Valine is a naturally occurring amino acid.



Valine

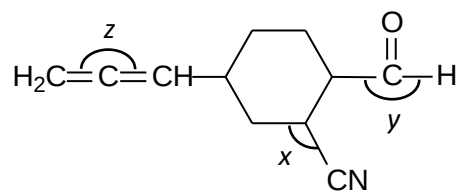


3-aminopropanoic acid

Which statement about valine and 3-aminopropanoic acid is correct?

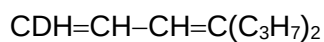
- A Both compounds are able to form zwitterions.
- B Both compounds are able to react with ethanoic acid to give amide.
- C Both compounds are able to react with NaBH_4 in methanol to give alcohol.
- D Both compounds are soluble in water due to the formation of intermolecular hydrogen bonding.

20 Which of the following statements are correct about compound **U**?



compound **U**

- 1 The bond angles in compound **U** increase in the order $x < y < z$.
 - 2 There is only one sp hybridised carbon atom in compound **U**.
 - 3 There are 26 σ and 5 π bonds.
 - 4 Compound **U** is planar.
- A 1 and 2 only
 B 1 and 3 only
 C 1, 2 and 4 only
 D 2, 3 and 4 only
- 21 Deuterium (D or ^2H) is a heavy isotope of hydrogen. A deuterio-hydrocarbon has the part structural formula shown below.



What is the total number of isomers with the above part-structural formula?

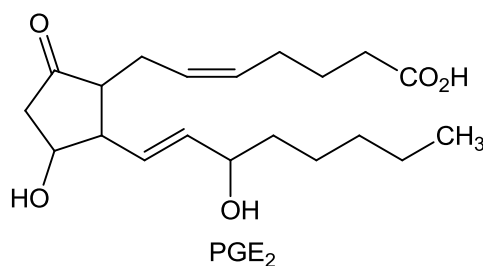
- A 2
 B 4
 C 8
 D 10

- 22 The molecule of benzene, C_6H_6 is a regular hexagon in which the π electrons are described as delocalised.

Which of the following statements support this structure?

- 1 Benzene is a good conductor of electricity.
 - 2 Addition reactions of benzene take place more easily than substitution.
 - 3 All C–C bonds lengths in benzene are intermediate between C–C bond in an alkane and C=C bond in an alkene.
 - 4 The enthalpy change of hydrogenation of benzene is less exothermic than that predicted for cyclohexatriene.
- A 1 and 3 only
 B 3 and 4 only
 C 1, 2 and 3 only
 D 1, 2 and 4 only

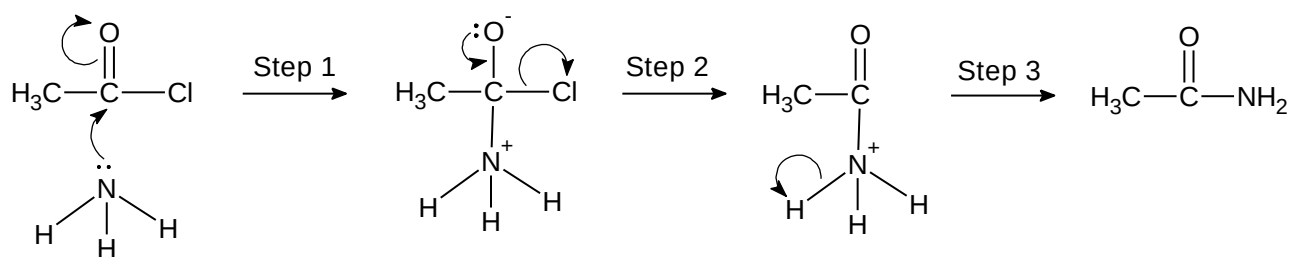
- 23 PGE_2 is a prostaglandin that has pharmacological activity.



Which of the following is incorrect?

	Reducing agent	Number of hydrogen atoms incorporated per molecule of PGE_2
A	H_2 / Ni	6
B	Na in ethanol	8
C	$NaBH_4$ in methanol	6
D	$LiAlH_4$ in dry ether	4

24 A mechanism for the reaction between ammonia and ethanoyl chloride is given below.

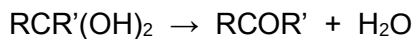


Which of the following statements are correct?

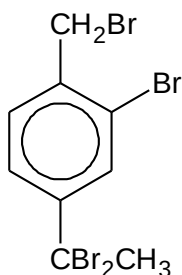
- 1 The ammonia behaves as a nucleophile.
- 2 The loss of proton in **step 3** is shown incorrectly.
- 3 The ammonia attacks an electron-deficient carbon atom.
- 4 The rate of reaction will increase if ethanoyl bromide is used.

- A** 1, 2, 3 and 4
B 1, 3 and 4 only
C 1 and 3 only
D 2 and 4 only

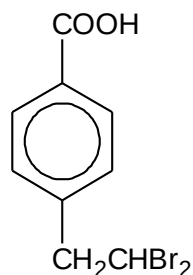
- 25** Diols formed on the same carbon atom are generally unstable and will undergo the following reaction.



Compounds **V** and **W** are both reacted with hot aqueous NaOH followed by dilute HCl at room temperature.



Compound **V**



Compound **W**

Which reagents could be used to distinguish the final products of **V** and **W**?

- 1 2,4-DNPH
- 2 Fehling's solution
- 3 Sodium carbonate
- 4 Hot acidified $\text{K}_2\text{Cr}_2\text{O}_7$
- 5 Iodine in aqueous NaOH

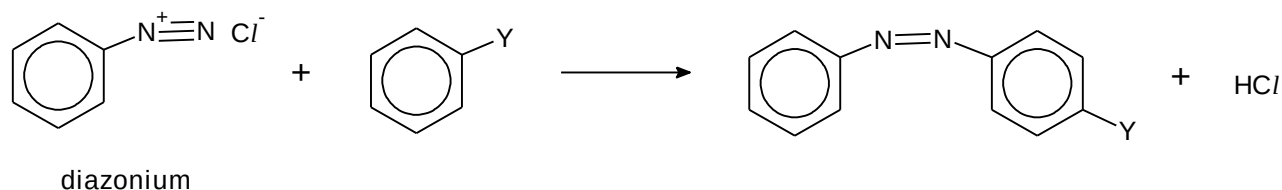
- A 1 and 2 only
- B 3 and 4 only
- C 1, 4 and 5 only
- D 2, 3 and 5 only

- 26** The ozone depletion potential (ODP) of a chemical compound is the relative amount of degradation to the ozone layer it can cause, with trichlorofluoromethane being fixed at an ODP of 1.0.

Which of the following compounds has an ODP of greater than 1.0?

- A methane
- B dichlorodifluoromethane
- C chlorotrifluoromethane
- D bromotrifluoromethane

- 27 Azo coupling is an organic reaction between a diazonium compound and another aromatic compound that produces an azo compound.



Which of the following correctly represents the type of reaction and the role of diazonium compound?

	Type of reaction	Role of diazonium
A	nucleophilic substitution	nucleophile
B	electrophilic substitution	electrophile
C	nucleophilic substitution	electrophile
D	electrophilic substitution	nucleophile

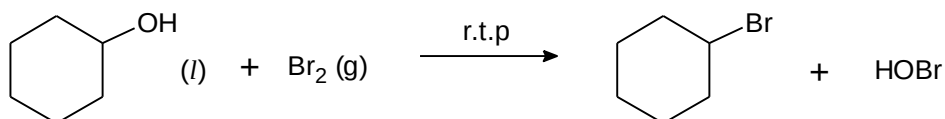
- 28 What is the order of increasing pH of 1 mol dm⁻³ of each organic compound?

- A CH₃CO₂Na, CH₃CO₂H, CF₃CO₂H, HOCH₂CHO
 B CH₃CO₂Na, HOCH₂CHO, CH₃CO₂H, CF₃CO₂H
 C CF₃CO₂H, CH₃CO₂H, HOCH₂CHO, CH₃CO₂Na
 D CF₃CO₂H, CH₃CO₂H, CH₃CO₂Na, HOCH₂CHO

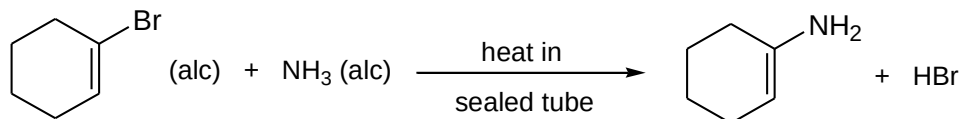
- 29 Which reaction gives the best yield of products?

[(alc) indicates an alcoholic solution.]

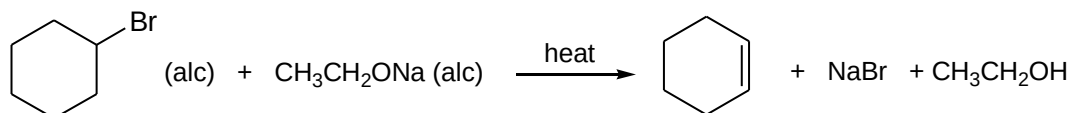
A



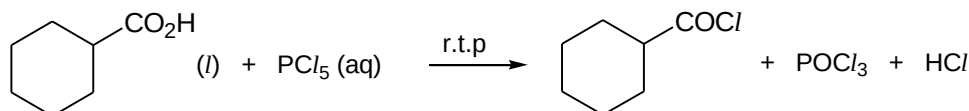
B



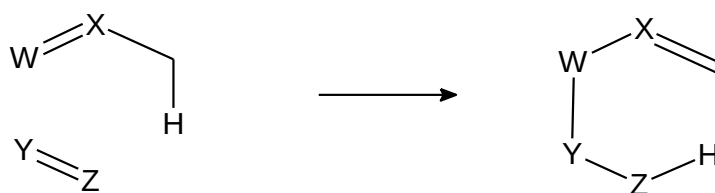
C



D



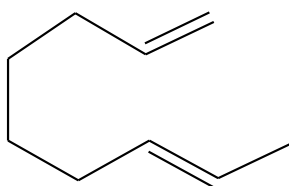
- 30 The Alder-ene reaction is a reaction between two alkenes and is capable of forming new carbon-carbon bonds. An example of the Alder-ene reaction is given below:



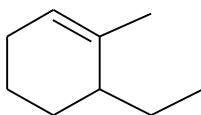
Note: W, X, Y & Z are carbon atoms and other hydrogens are not shown

The Alder-ene reaction typically produces many products as the roles of the two alkenes can be exchanged.

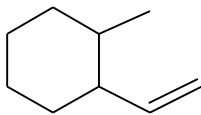
Which compound will **not** be formed when the following compound undergoes Alder-ene reaction?



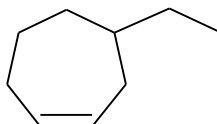
A



B



C



D

